

Real-time Disaster Search & Rescue System utilizing Semantic Segmentation and Object Recognition onboard Fixed-wing UAV

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Traditional search-and-rescue (SAR) operations rely heavily on manual aerial interpretation and often fail under low-visibility or communication-denied conditions. This project aims to address the engineering challenge of performing multimodal survivor detection under extreme pixel sparsity and strict edge power constraints, without reliance on cloud infrastructure.

A fully onboard artificial intelligence system was engineered on an NVIDIA Jetson Orin NX platform with a custom TensorRT-accelerated inference pipeline. The vision subsystem integrates a modified YOLOv11 detector optimized for sub-10-pixel aerial human targets and a custom Swin Transformer-based terrain segmentation model. Both networks were restructured using FP16 quantization, layer fusion, and pipeline parallelization to maximize performance per watt and minimize latency. Visual inference was integrated with a four-microphone array for voice direction-of-arrival estimation and a wireless network interface for RF presence detection through a custom weighted multimodal fusion architecture.

Engineering validation emphasized throughput, latency, and power efficiency under full multimodal load. The system achieved a 95th percentile throughput of 156 frames per second while maintaining a 95th percentile power draw of 25.0 W, satisfying real-time flight constraints. Integrated mission testing confirmed synchronized cross-modal detection and successful human identification at 200 feet under both visible and partially occluded conditions. This test achieved a detection rate 2.6 times greater than that of a single-modality (vision-only) test.

By demonstrating that transformer-based segmentation, real-time object detection, and acoustic/RF sensing can operate concurrently within strict edge power budgets, this work establishes a scalable systems-engineering framework for resilient disaster-response UAV platforms.